Introduction

The geometric distribution counts the number of Bernoulli trials required to obtain the first success. It is the discrete analogue of the exponential waiting-time distribution and shares the memoryless property: the probability of further trials to success does not depend on how many have already been tried.

Prerequisites

Bernoulli trials.

Theory

Two conventions exist. Let X be the number of Bernoulli(p) trials *up to and including* the first success:

$$P(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

Alternatively, Y is the number of failures *before* the first success:

$$P(Y = k) = (1-p)^{k}p, \qquad k = 0, 1, 2, \ldots$$

R uses the second convention (`rgeom`, `dgeom` return Y, i.e., failures before first success).

Moments for the first convention (X = trials until first success):

- $E[X] = 1/p$, $\mathrm{Var}(X) = (1-p)/p^2$.

For Y (R's convention): $E[Y] = (1-p)/p$, $\mathrm{Var}(Y) = (1-p)/p^2$.

Memoryless property: $P(X > m + n \mid X > m) = P(X > n)$. Having waited m trials without success does not change the distribution of further waiting time.

Assumptions

Independent Bernoulli trials with constant probability p.

R Implementation

```r
p <- 0.3

# R's convention: Y = failures before first success
Y <- rgeom(1e5, p)
c(mean_Y = mean(Y), theoretical = (1 - p) / p)

# Number of trials to first success X = Y + 1
X <- Y + 1
c(mean_X = mean(X), theoretical = 1 / p)

# PMF comparison
k <- 0:15
data.frame(k, pmf_Y = dgeom(k, p))

# Memoryless: P(Y > 5 + 3 | Y > 5) = P(Y > 3)
Y_past_5 <- Y[Y > 5]
c(conditional = mean(Y_past_5 > 5 + 3),
  unconditional = mean(Y > 3))
```

Output & Results

```
   mean_Y theoretical
    2.334       2.333

   mean_X theoretical
    3.334       3.333

    k      pmf_Y
1   0      0.3000
2   1      0.2100
3   2      0.1470
...

conditional  unconditional
     0.3441         0.3430
```

The memoryless property holds empirically: conditional probability of further 3 failures, given already 5 failures, equals the unconditional probability of at least 3 failures.

Interpretation

The geometric distribution is the baseline model for "time to event" in the discrete case: cycles to pregnancy in IVF, attempts before a successful procedure, trials before a rare positive test. Memorylessness is a strong assumption that often fails in real data where fatigue or learning changes p over time.

Practical Tips

- Check which convention (trials vs. failures) is expected when comparing to textbook formulas.
- Memorylessness is testable empirically: compare the distribution of residual trials given survival to each time.
- For non-memoryless discrete waiting times, the negative binomial (shape > 1) captures over-dispersion relative to geometric.
- The geometric converges to the exponential when discretising time with fine intervals.
- In clinical or biological settings, censoring is common; naive `rgeom` ignores right-censoring – use survival methods for realistic data.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots